Understanding TOF Timing

Timing is everything, especially with TOFs. This manual explains it.

1. Understanding the correlation between timing and data

TOF data is very closely related to the TOF timing. Therefore you may want to read the manual on TOF data before reading this manual.
Tofwerk's TOFs are especially suited for monitoring fast processes, thereby recording multidimensional data. This means that the data recording process consists of several nested loops of data. The lowest level of data consists of the time-of-flight measurement of the ions. Therefore, at least in the case of the TDC, a sample represents the number of ions with a specific time-of-flight[1]. The samples are usually clocked by a clock in the DAQ hardware[2]. The samples of one TOF extraction form the next level data structure, a TOF histogram[3] also called waveform. The TOF extraction is clocked by a trigger issued by the timing generator (TIGER). A process can be monitored by a block of extractions. A block can be triggered in three different ways: (1) by overflow, which means the previous block was completed, (2) by a dedicated signal from the TIGER, or (3) by an external block trigger. Several blocks can form a cube[4]. A cube is used to record the temporal evolvement of blocks. Currently, cubes do not have dedicated triggers. This means that if several writes are recorded, the next write starts immediately after the previous write is completed (overflow trigger method). A series of cubes is a tetra, a 4-dimensional data structure. It could be that this nesting will be continued and that the software will offer to record pentas, a 5-dimensional data structure.
A RUN is a complete TOF data acquisition cycle. Between RUNs the hardware is reinitialized, which means data within a run are recorded with the same settings, whereas data from different runs may have different settings and are saved in individual files. A RUN corresponds to the largest data structure. This means a RUN currently corresponds to a tetra, except if the number of cubes is equal to 1 (a single cube is recorded) then a RUN corresponds to a cube.

data	triggered by	trigger method	period	pixel of	constitutes
sample	DAQ clock	DAQ intern	0.1 - 1 ns	Dimension 1	-
waveform	DAQ trigger	TIGER	10 -50 us	Dimension 2	Dimension 1
block	block trigger	overflow/TIGER/external	1 ms	Dimension 3	Dimension 2
cube	-	overflow	1 s	Dimension 4	Dimension 3
tetra	START	overflow/external		-	Dimension 4

2. Averaging

Extractions (= waveforms) and blocks can be averaged. This allows for much more compact data if the maximum time resolution of one process is not needed. An averaged waveform is called "segment". An averaged block is called "buf". Samples, cubes and tetras cannot be averaged. Samples cannot be averaged because it would decrease TOF resolution. Cubes and tetras cannot be averaged because (1) so far nobody requested it, and (2) it would require reloading data from the hard disk into RAM which is better done off-line during data post processing.

[1] In the case of an ADC the situation is more complicated because a single ion can be registered in several neighbouring samples.
[2] For some DAQ hardware it is possible to override the internal clock and use an external clock for the sampling clock.
[3] Actually, time is not yet normalized to TOF in these histograms
[4] Several blocks can be stored in the PC memory and can then be written to disk – therefore the name „write".

Example: you may want to analyze a pulsed laser process (1 kHz = 1 ms) with a series of 10 TOF spectra of 20 us each and you may want to record a 1 h time trend in your data with 1 s resolution. This means in 1 s you can average the data of 1000 laser pulses. This cascade of three processes will yield 3D data.

Data dimension	meaning	name	period	contains	timing source
Dimension 1	TOF extraction	waveform	20 us	20000 samples	DAQ trigger
Dimension 2	laser pulse	block	1 ms	10 spectra	block trigger
	averaging	buf	1 ms	1000 pulses	none
Dimension 3	time trend	cube=write	1 s	3600 s	none

3. Understanding TIGER signals

The following table indicates the different levels of data recording and their signals.

data level	TOFMS name	TIGER signal	purpose	in / out	
sample	time bin	none			
waveform	extraction	master	TIGER internal only		T0
“		TTL1 & TTL2	TOF extractor	out	T6, T7
“		DAQ trigger	trigger for DAQ electronics	out	T4
“		TTL	blanking	out	T1
block	laser, chopper, etc	block in	external block trigger	in	T2
“		block out	block trigger	out	T3
“	laser, pulser	pulse out	trigger external experiment	out	T5
cube		-	overflow		
run		READY	DAQ is ready for next run	out	
		START	start run	in	
		BUSY	DAQ is busy	out	
		STOP	stop run	in	

Note that the RUN signals are processed by software, whereas all other triggers are usually processed by the TIGER without any software interaction.

4. Understanding timing modes

Different experiments use different hardware and so far it has not been possible to use the same TIGER program for all experiments. Therefore several timing modes have been implemented to cover different requirements:

mode	purpose
M0	generic mode
A1	simple mode
A2	external block trigger, has jitter
A3	external block trigger, needs additional cable
A4	not yet functional
G4	internal block trigger but waits for ADC
G5	not yet functional
G6	external block trigger and waits for ADC
I1	external block trigger and waits for ADC, imaging
I2	external block trigger and waits for TDC, imaging

A main reason for the number of timing modes is the limited capabilities of the ADC data acquisition electronics. The ADC in averager mode cannot record data and transfer data to the PC at the same time. This means there are gaps in the data. In many cases these gaps can be managed in software in a way that they do not matter at all. In other cases they do matter and the TIGER minimizes the gaps by waiting for a READY hardware signal from the ADC in order to continue recording data as immediately as possible.

4.1. Modes M0: Continuous

M0 is an extension of A1. M0 is called a “continuous mode” because it continuously records TOF extractions. It does not need a block dimension. These modes can be used when a continuous series of mass spectra should be taken. In these modes it is best to have NbrSegments = 1, which means each spectrum is separately downloaded to the PC, leading to a short data gap after each spectrum. This means there is no synchronization it a fast external process like a laser, a pulser, a chopper, necessary. However, it is still possible to issue block pulses. It is also possible to record a block of spectra for each block pulse in cases where the DAQ can be inhibited by a TIGER signal. The problem is that if the next block is issued before the DAQ is ready for new data (data download still going on), the TIGER will loose synchronization with the DAQ electronics.

4.2. Mode A2: External Block Trigger

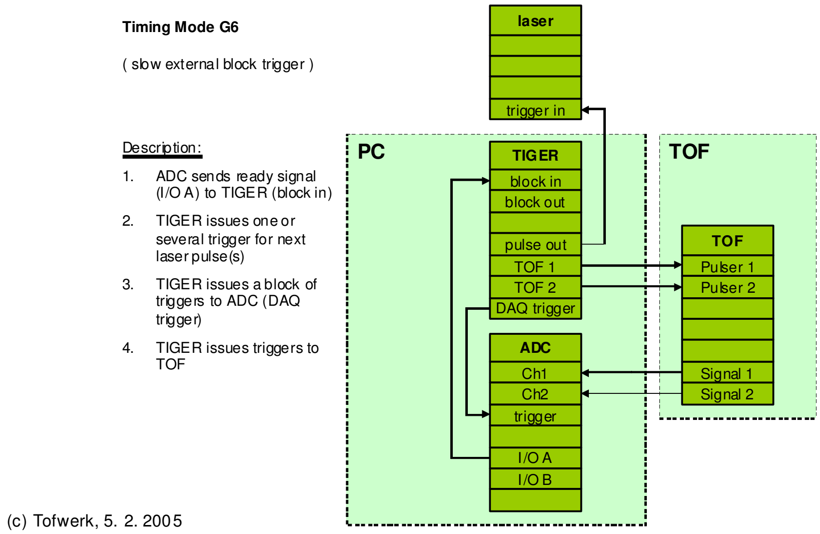
A2 is a mode that should no longer be used since better modes have been developed in the mean time. It allows for recording a series of spectra (in order to monitor a fast process) after an external block trigger is received. A2 synchronizes the (continuously extracting) TOF with an external signal. However, the jitter in synchronization is 4 TOF extractions instead of only one TOF extraction.

4.3. Mode A3: External Block Trigger

A3 is superseding A2. It allows for recording a series of spectra (in order to monitor a fast process) after an external block trigger is received. If the external trigger is unsynchronized with the (continuously extracting) TOF, the jitter is one TOF cycle. For less jitter, please allow a TIGER block trigger to control the external process. A3 is stopped during data download and is restarted after receiving a software READY from the DAQ. This means that A3 is susceptible to Windows time delays. Use Mode G6 for a software independent timing mode.

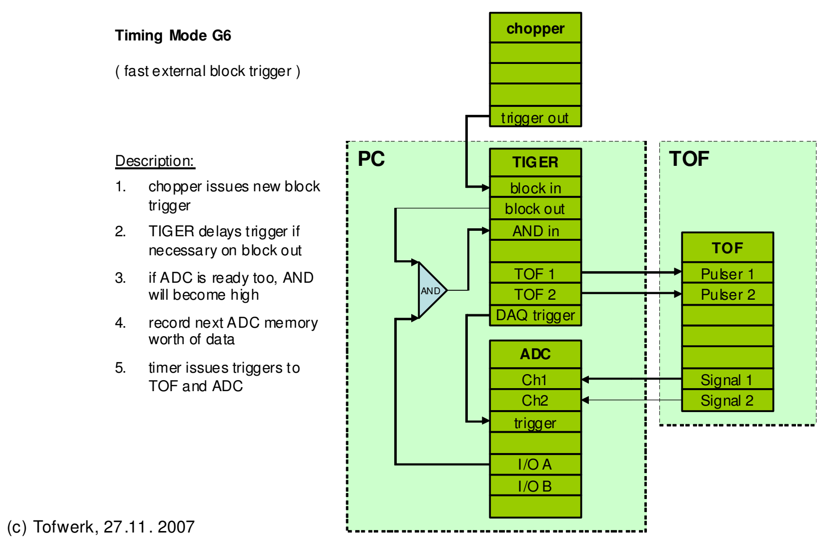
4.4. Mode G4: TIGER Block Trigger

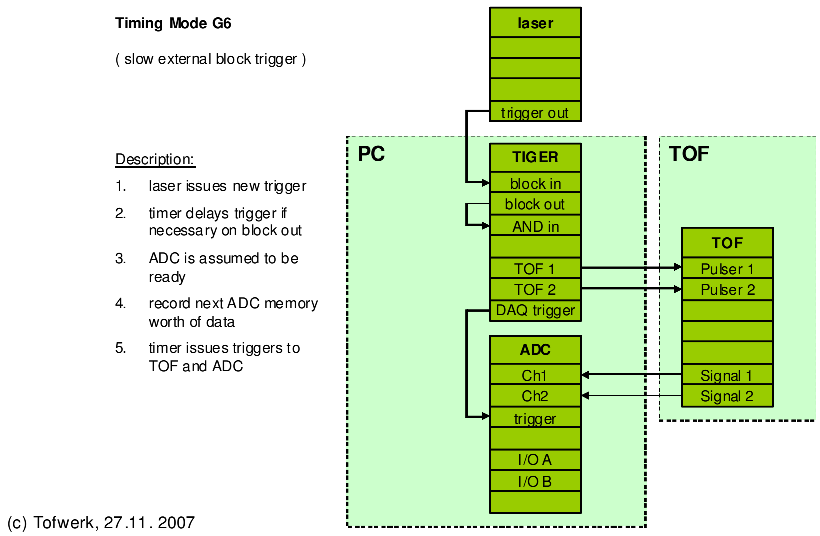
When the TIGER issues the block trigger, synchronization with the TOF is much better. The jitter is in the order of ns instead of tenths of us. If using an ADC in averager mode, use G4 if ever possible. During download G4 will stop issuing block triggers until the DAQ sends a hardware signal that it is ready to record again. Upon that signal G4 will restart issuing block triggers.



4.5. *Mode G6: External Block Trigger*

G6 is an improved version of A3. G6 is not stopped during DAQ data download. Instead it waits for a hardware READY signal from the DAQ and then immediately continues the measurement. This means G6 is completely independent on Windows timing. If external block triggers are slow and there is no risk that data transfer is not finished when next block trigger is issued, the AND gate can be omitted.





4.6. Mode I1: External Block Trigger Imaging

I1 is similar to G6. However, it has two nested block schemes which allow for selecting lines and rows in a raster scan image.

4.7. Mode I2: External Block Trigger Imaging

I2 is similar to I1 but for DAQ systems with simultaneous recording/downloading. TDCs and some ADCs allow for simultaneous downloading and recording. This is a big advantage in imaging. Images can be recorded with much higher duty cycle, not restricted by data transfer rates.